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Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 5	
TITLE: Collection Classes and StringBuffer	

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

1. Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.

Programme

```java
import java.util.ArrayList;

public class Assignment5 {

    Run | Debug
    public static void main(String[] args) {

        // ArrayList to store tasks
        ArrayList<String> tasks = new ArrayList<>();

        tasks.add(e: "Complete Java Assignment");
        tasks.add(e: "Study Wrapper Classes");
        tasks.add(e: "Submit Lab Assignment");

        // StringBuffer to display tasks
        StringBuffer sb = new StringBuffer();

        sb.append(str: "To-Do List:\n");

        for (int i = 0; i < tasks.size(); i++) {
            sb.append((i + 1) + ". " + tasks.get(i) + "\n");
        }

        System.out.println(sb);
    }
}
```

Output

```
PS C:\Users\RIKISHA\PROGRAMMING\Java>  c:; cd 'c:\Users\RIKISHA\PROGRAMMING\Java'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:49713' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\RIKISHA\PROGRAMMING\Java\bin' 'Assignment5'
To-Do List:
1. Complete Java Assignment
2. Study Wrapper Classes
3. Submit Lab Assignment

PS C:\Users\RIKISHA\PROGRAMMING\Java>
```

2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

Programme

```java
import java.util.ArrayList;

public class Assignment5 {

    Run | Debug
    public static void main(String[] args) {

        // ArrayList to store registered courses
        ArrayList<String> courses = new ArrayList<>();

        // Add courses
        courses.add(e: "Java Programming");
        courses.add(e: "Data Structures");
        courses.add(e: "Database Management");

        // Remove one course
        courses.remove(o: "Data Structures");

        // Display courses using StringBuffer
        StringBuffer sb = new StringBuffer();

        sb.append(str: "Registered Courses:\n");

        for (String course : courses) {
            sb.append(course + "\n");
        }

        System.out.println(sb);
    }
}
```

Output

```
PS C:\Users\RIKISHA\PROGRAMMING\Java> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:57160' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\RIKISHA\PROGRAMMING\Java\bin' 'Assignment5'
Registered Courses:
Java Programming
Database Management

PS C:\Users\RIKISHA\PROGRAMMING\Java>
```

Conclusion

The program successfully demonstrated the use of ArrayList, Vector, and StringBuffer. ArrayList and Vector were used for dynamic data storage and management, while StringBuffer was used for efficient string manipulation. This shows how Java Collection classes and StringBuffer simplify handling dynamic data and mutable strings.